

What's New in Modern .NET Library APIs

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Downloads

<https://github.com/JasonBock/WhatsNewInNETLibraryAPIs>

<https://github.com/JasonBock/Presentations>

Overview

- Focus
- .NET Features
- Call To Action

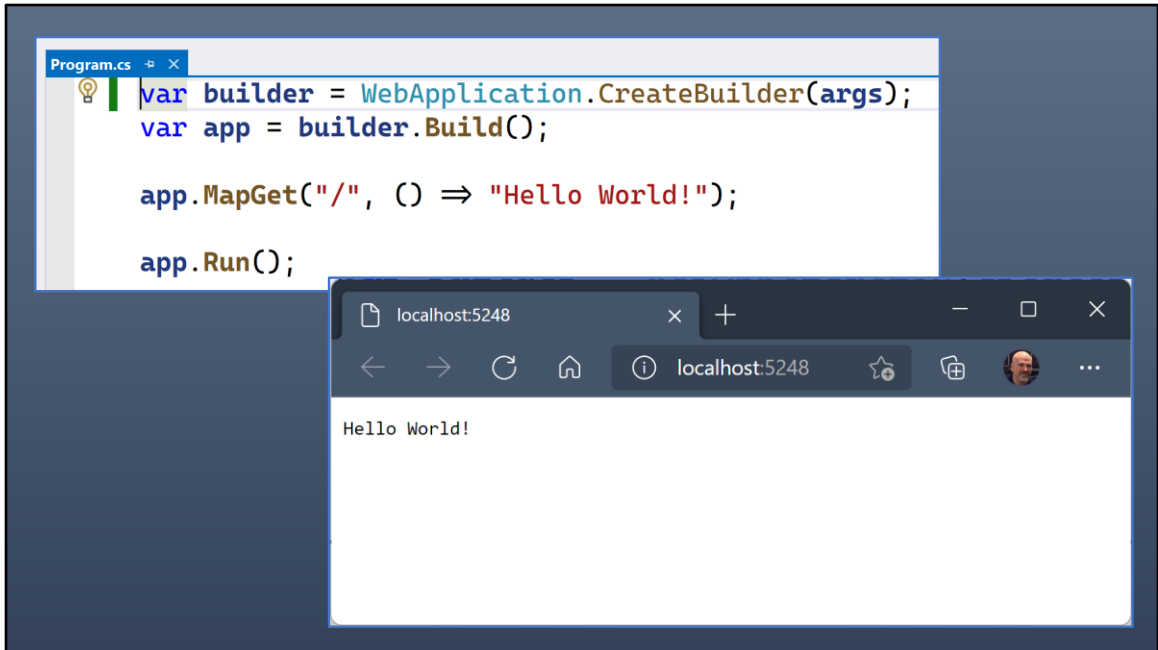
Remember...

<https://github.com/JasonBock/WhatsNewInNETLibraryAPIs>

<https://github.com/JasonBock/Presentations>

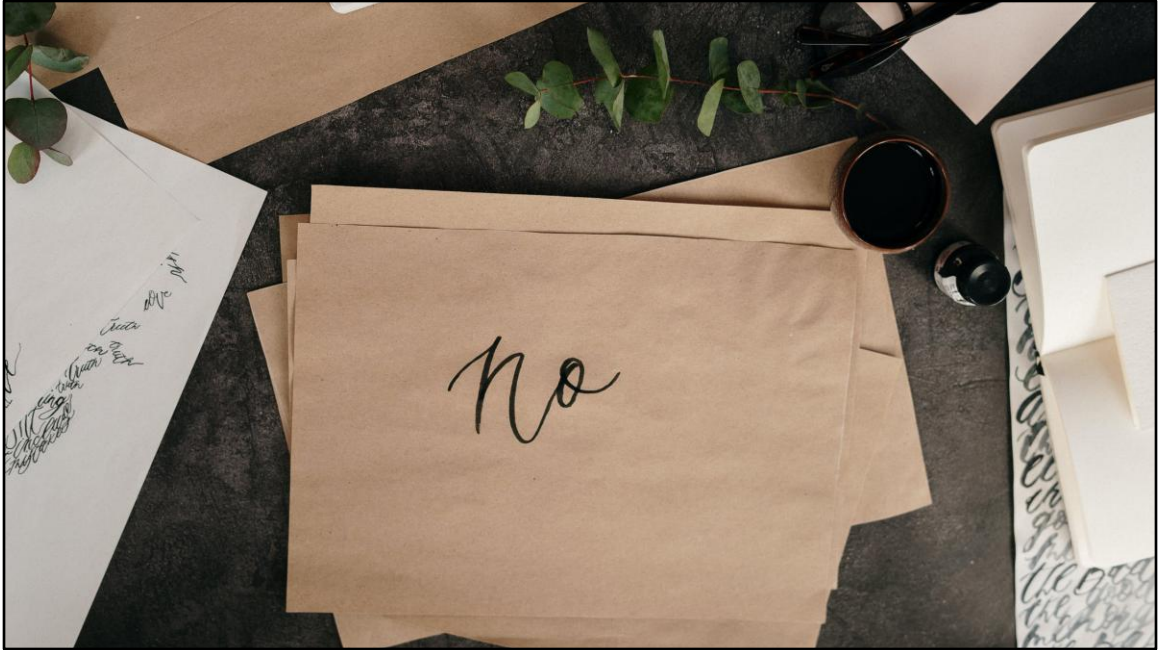
Focus

What's New in Modern .NET Library APIs



Let's level-set what we're going to be talking about. When developers hear "API", they may think that we're talking about "REST APIs". Or, if they use .NET, they may have heard of "Minimal APIs" in .NET 6. Either way, they might think I'm going to talk about one....or both.

<https://docs.microsoft.com/en-us/aspnet/core/fundamentals/minimal-apis>



That's not it. Those are definitely worth discussing, but that's not the focus of this talk.

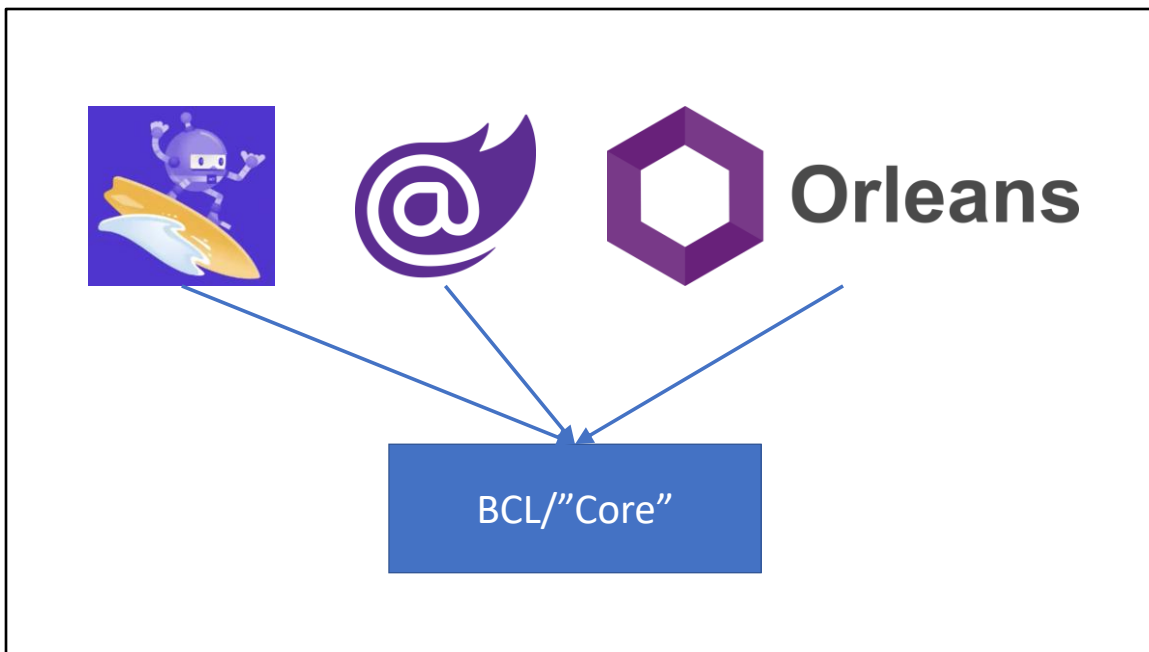
<https://www.pexels.com/photo/brown-paper-with-handwritten-text-3826681/>

An application programming interface (API) is a connection between computers or between computer programs. It is a type of software interface, offering a service to other pieces of software. A document or standard that describes how to build or use such a connection or interface is called an API specification. A computer system that meets this standard is said to implement or expose an API. The term API may refer either to the specification or to the implementation.

This is a somewhat formal definition of an API.

<https://en.wikipedia.org/wiki/API>

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There are a bunch of frameworks, like MAUI, Blazor, Orleans, etc. that have their own APIs. What I'm referring to in this talk is the "Base Class Library". These are the core types.

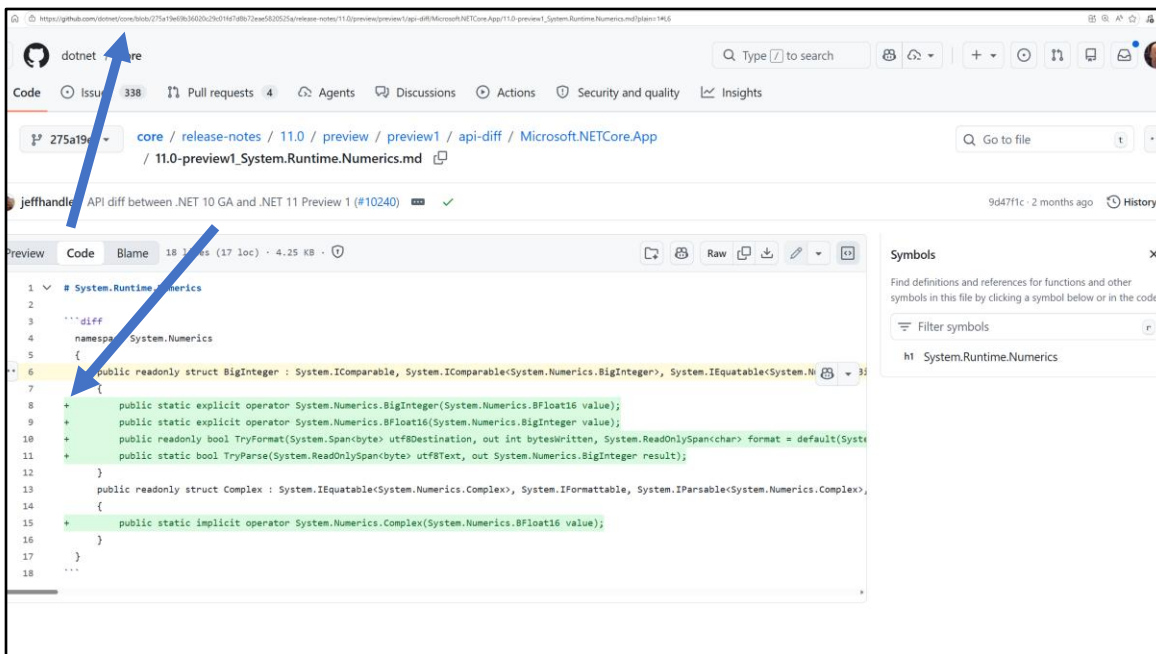
What are the "core" types? The "Base Class Library" definition was always kind of nebulous. .NET Standard provided a "standard", and now .NET 5 and beyond provides that "standard".

The screenshot shows the Microsoft Learn .NET API browser interface. At the top, there is a navigation bar with links for Learn, Documentation, Training & Labs, Q&A, and Topics. Below this, a secondary navigation bar includes Languages, Features, Workloads, APIs, Troubleshooting, and Resources. A blue button labeled 'Download .NET' is in the top right corner. The main heading is '.NET API browser'. Below it, a welcome message states: 'Welcome to the .NET API browser – your one-stop shop for all .NET-based APIs from Microsoft. Start searching for any managed APIs by typing in the box below.' A link to the 'dotnet/dotnet-api-docs' repo is provided for feedback. A search filter 'API: All APIs' is active. The search input field contains 'BigInteger'. Below the search bar, the section '.NET API Reference' contains a table with the following data:

Name	Description
Java.Math.BigInteger Class	Immutable arbitrary-precision integers.
System.Numerics.BigInteger Struct	Represents an arbitrarily large signed integer.
Orleans.Serialization.Codecs.BigInteger Codec Class	Serializer for BigInteger.
Java.Math.BigInteger(BigInteger)	Translate a BigInteger containing the two's complement binary representation of a BigInteger into a

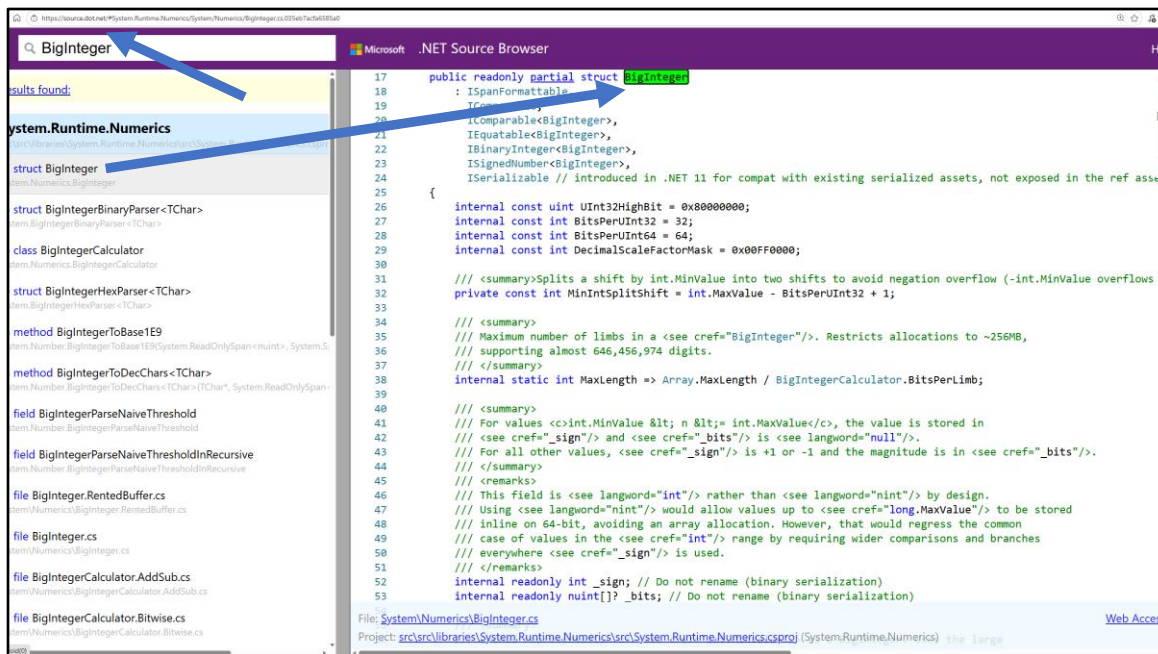
So what defines what's in each version? You can use the API browser, and filter it to whatever version you care about.

<https://learn.microsoft.com/en-us/dotnet/api/>



Another place is to go to the dotnet/core repository, and look at the diff files produced by Microsoft. They're under the "release-notes" folder. It's not the greatest way, but you find everything that's changed.

<https://github.com/dotnet/core/blob/main/release-notes>



And if you're curious to see how it's defined, you can use the Source Index browser!

<https://source.dot.net/>



One thing to keep in mind before I do the demos. Some of these additions won't matter for you, where some of them will. The relevance of the feature is highly dependent on what you do in your code. So, don't be surprised if only a couple grab your attention. Then again, you never know where you may need some of these.

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September 10th, 2025 91 reactions

Performance Improvements in .NET 10



Stephen Toub - MSFT
Partner Software Engineer



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My kids love "Frozen". They can sing every word, re-enact every scene, and provide detailed notes on the proper sparkle of Elsa's ice dress. I've seen the movie more times than I can recount, to the point where, if you've seen me do any live coding, you've probably seen my subconscious incorporate an Arendelle reference or two. After so many viewings, I began paying closer attention to the details, like how at the very beginning of the film the ice harvesters are singing a song that subtly foreshadows the story's central conflicts, the characters' journeys, and even the key to resolving the climax. I'm slightly ashamed to admit I didn't comprehend this connection until viewing number ten or so, at which point I also realized I had no idea if this ice harvesting was actually "a thing" or if it was just a clever vehicle for Disney to spin a yarn. Turns out, as I subsequently researched, it's quite real.

In the 19th century, before refrigeration, ice was an incredibly valuable commodity. Winters in the northern United States turned ponds and lakes into seasonal gold mines. The most successful operations ran with precision: workers cleared snow from the surface so the ice would grow thicker and stronger, and they scored the surface into perfect rectangles using horse-drawn plows, turning the lake into a frozen checkerboard. Once the grid was cut, teams with long saws worked to free uniform blocks weighing several hundred pounds each. These blocks were floated along channels of open water toward the shore, at which point men with poles levered the blocks up ramps and hauled them into storage. Basically, what the movie shows.

The storage itself was an art. Massive wooden ice houses, sometimes holding tens of thousands of tons, were lined with insulation, typically straw. Done well, this insulation could keep the ice solid for months, even through summer heat. Done poorly, you would open the doors to slush. And for those moving ice over long distances, typically by ship, every degree, every crack in the insulation, even a tiny leak in transit meant more melting and more loss.

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But even if none of these grab your attention, you should still consider upgrading to the latest version. Since .NET Core, there has been a lot of engineering effort put into making the runtime and core types better from a perf perspective. The latest article from Taub illustrates the continuation of this journey. Simply upgrading to the new version has a big potential of improving the performance of your applications.

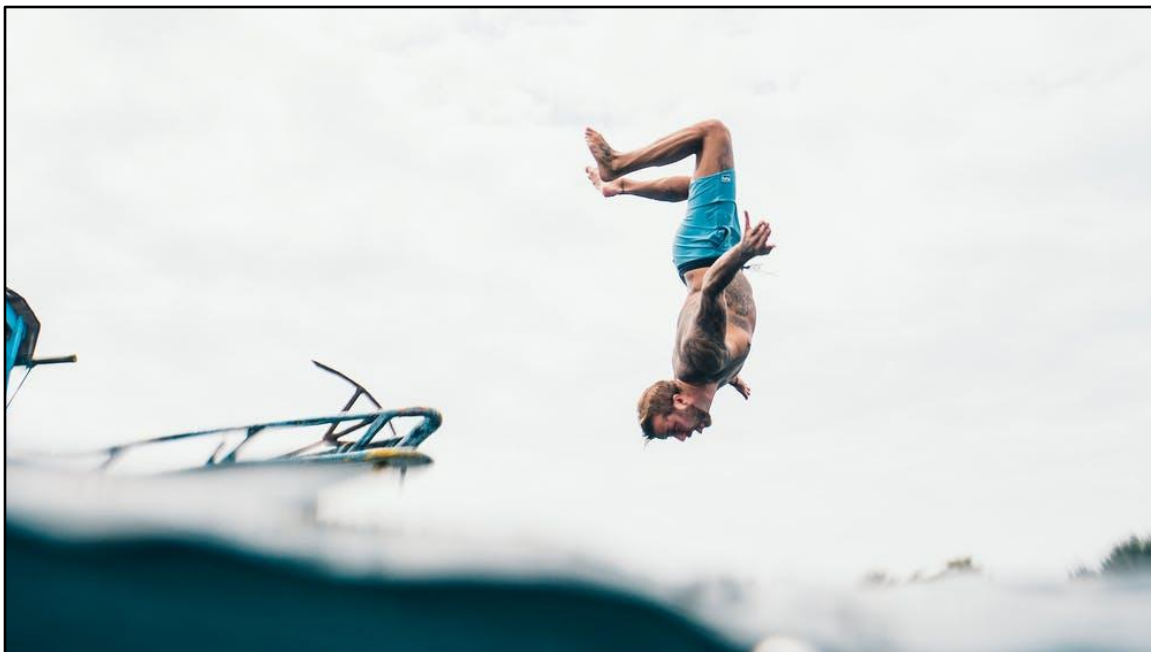
<https://devblogs.microsoft.com/dotnet/performance-improvements-in-net-10/>

Demo: .NET Features

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Call To Action

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Dive in! Each new version of .NET has a bunch of new APIs that you may want to use, and the entire .NET API surface is large. You may find some new types or methods that you can use for your own applications.

<https://www.pexels.com/photo/man-wearing-blue-shorts-about-to-dive-on-body-of-water-1098365/>

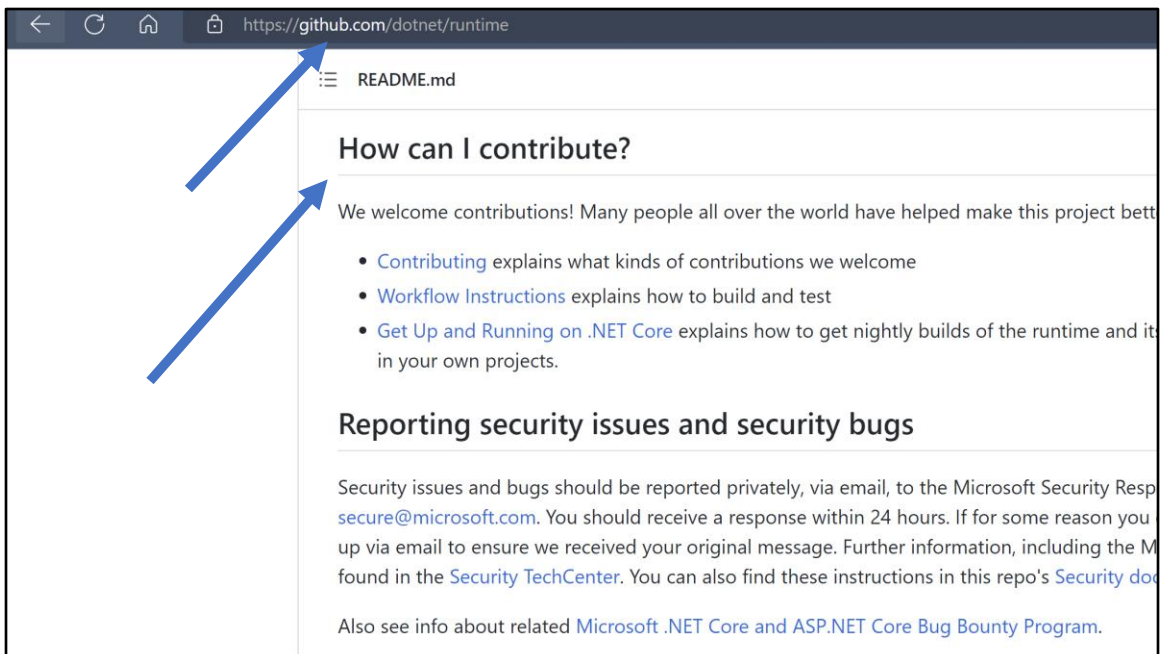


Explore what's currently there. There's so much there that's been added from previous versions that you may not be aware of. Go back to the resources I showed before and see what's there.

<https://unsplash.com/photos/UVyOfx3v0Ls>

Demo: Unsafe.As()

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Finally, if you want to contribute, either by suggesting new features, or by doing the work, visit the .NET runtime repository.

<https://github.com/dotnet/runtime>

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- References in the notes on this slide

References

* Sources

* [Official Start
Page](<https://dotnet.microsoft.com/>)
* [.NET
Runtime](<https://github.com/dotnet/runtime>)
* [Source Index](<https://source.dot.net/>)
* [.NET API
Browser](<https://docs.microsoft.com/en-us/dotnet/api/>)
* [.NET Release
Notes]([https://github.com/dotnet/core/tree/main/release-
notes](https://github.com/dotnet/core/tree/main/release-notes))
* [Themes of .NET](<https://themesof.net/>)